

Project Design Phase Solution Architecture

Date	19 July 2026
Team ID	
Project Name	AI-Driven Emotion Detection & Personalized Learning Support Platform
Maximum Marks	4 Marks

Solution Architecture:

The proposed solution is an AI-powered web application that analyzes students' text inputs to detect emotions using BERT and BiLSTM models. Based on the detected emotion, the system generates personalized learning guidance through the Gemini API. User interactions are stored in CSV logs and visualized in a Streamlit analytics dashboard. The architecture ensures accurate emotion detection, scalable deployment, and an interactive learning experience.

Example - Solution Architecture Diagram:

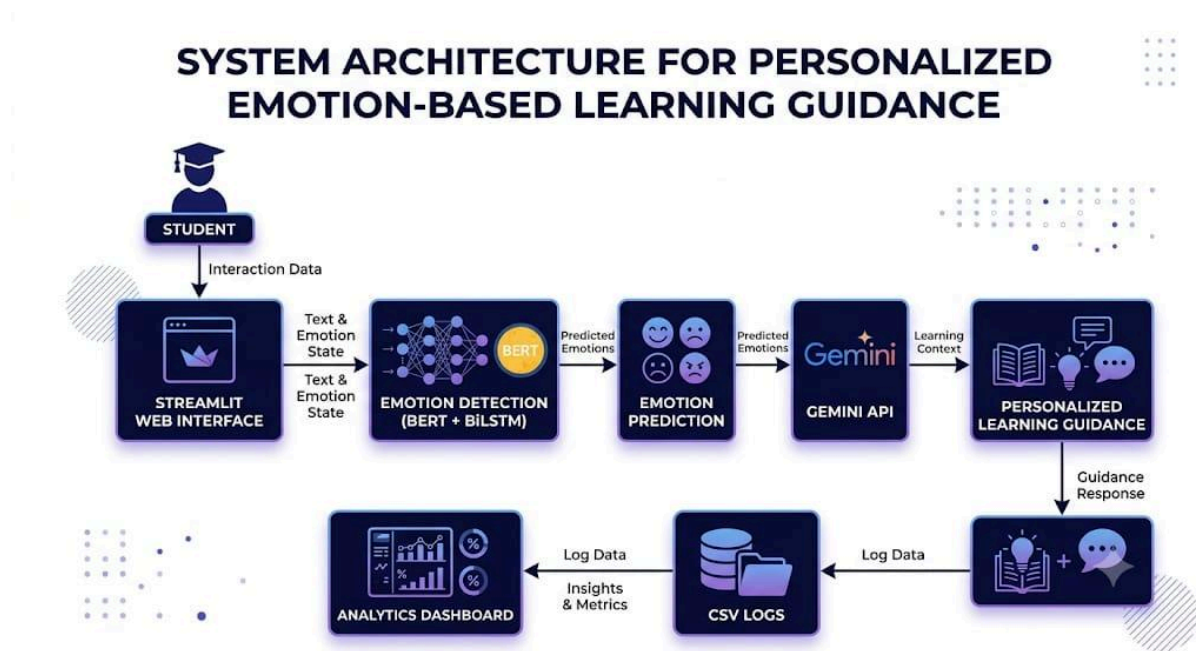


Figure 1: Architecture of the emotion based learning guidance

Reference:

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>